

Research Statement

Trung Dang

A central theme of my past and current research has been designing practical algorithms under *information uncertainty, strategic behavior, and/or data corruption*. This captures many real-world scenarios, such as auction design with strategic participants, decision-making under noisy machine-learned data, etc. In these practical scenarios, a **simple algorithm** that uses limited information is often preferred over complex algorithms. It is therefore also important to understand the theoretical guarantees of these simple algorithms, which is my main research focus. Some of my past work within this theme is:

- In [CCDH⁺25], we provided the **first approximation** for combinatorial revenue maximization, achieved via a **simple item pricing mechanism** that can be computed in polynomial time. The main difficulty of this problem lies in the sensitivity of each buyer’s preference under different possible item prices, which makes it difficult to protect some high-value items for later buyers. We resolved this via a careful randomization of the item pricing, turning a much weaker ex ante guarantee on “the number of items sold to each buyer” to the much stronger ex post guarantee that we want.
- In [CCD25], we provided and/or improved the approximation ratio of a **combinatorial generalization** of Pandora’s Box under many different feasibility constraints. We achieved this by relating the performance of a **simple algorithm class** in this setting to that of *threshold algorithms* in prophet inequality, which are known to have good competitive ratios.
- In [ADMP⁺26], we essentially resolved the problem of “selecting the largest random variable only knowing the noisy initializations”. We show that a **very simple strategy** achieves a constant approximation in a certain regime, while no algorithms can achieve good approximations outside of such a regime. This is in contrast to our first attempt of trying to learn the underlying distributions and then employ the optimal algorithm knowing the distributions, which proved difficult to analyze.

Outside of algorithms under uncertainty, I am also generally interested in algorithmic statistics, including studying the optimality and robustness of algorithms in different statistical settings. For example, in [DLSV23], we show that the simple median-of-means algorithm for mean estimation enjoys beyond-worst-case guarantees: within a well-defined neighborhood of any distribution, no other algorithms can beat the error rate of the median-of-means algorithm. In [DH25], we propose a new model of adversarial data corruption that has applications in parallel computing, and also provide near-optimal simple algorithms for computing the maximum element under this model. My interest in this field, particularly the optimality and robustness of simple algorithms like median-of-means and robust max selection, serves as a natural extension of my core work in algorithmic economics. Both fields share the common challenge of designing simple, dependable mechanisms that perform well despite real-world complexities, be they strategic agents or adversarial data corruption.

References

Cited papers

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- [CCD25] Shuchi Chawla, Dimitris Christou, and Trung Dang. Commitment gap via correlation gap. In *International Conference on Web and Internet Economics*, 2025.
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- [DH25] Trung Dang and Zhiyi Huang. Robust max selection. In *2025 IEEE International Symposium on Information Theory (ISIT)*, pages 1–6. IEEE, 2025.
- [DLSV23] Trung Dang, Jasper Lee, Maoyuan Song, and Paul Valiant. Optimality in mean estimation: beyond worst-case, beyond sub-gaussian, and beyond $1 + \alpha$ moments. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 4150–4176. Curran Associates, Inc., 2023. URL: https://proceedings.neurips.cc/paper_files/paper/2023/file/0cddb777d3441326544e21b67f41bdc8-Paper-Conference.pdf.

Supplementary papers

- [CDHW25] Shuchi Chawla, Trung Dang, Zhiyi Huang, and Yifan Wang. Multi-unit combinatorial prophet inequalities. In *International Conference on Web and Internet Economics*, 2025.
- [DMVW23] Trung Dang, Walter McKelvie, Paul Valiant, and Hongao Wang. Improving pearson’s chi-squared test: hypothesis testing of distributions—optimally. *arXiv preprint arXiv:2310.09408*, 2023.
- [LDWL21] Zitao Li, Trung Dang, Tianhao Wang, and Ninghui Li. Mgd: a utility metric for private data publication. In *Proceedings of the 8th International Conference on Networking, Systems and Security, NSysS ’21*, pages 106–119, Cox’s Bazar, Bangladesh. Association for Computing Machinery, 2021. ISBN: 9781450387378. DOI: 10.1145/3491371.3491385. URL: <https://doi.org/10.1145/3491371.3491385>.